

Co-Architecture and the Detection Method

How the human + AI discipline made the vulnerability class visible

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Paper 2 of 4 in the GIC series.

The Discipline of Co-Architecture

Co-Architecture is the structural discipline that allowed the NATV vulnerability class to be discovered. It is not a tool, a scanner, or a workflow; it is a governed partnership between human reasoning and AI reasoning, each operating inside a shared constraint surface. Humans define boundaries, intent, continuity requirements, and governance rules. AI evaluates structure, propagation, invariants, and state transitions at a scale no human can hold in working memory. When these two reasoning modes operate together inside a governed substrate, the system becomes capable of revealing architectural truth that neither side could expose alone. This is the core of Co-Architecture: a dual-operator method where human constraint fields and AI structural fields align to evaluate whether a system is actually governed.

The discipline works because humans and AI reason in fundamentally different ways. Humans understand purpose, context, and governance - they know why a boundary exists and what must never be allowed to cross it. AI understands structure. It can trace identity propagation across thousands of operations, evaluate constraint surfaces, and detect continuity fractures that would be invisible to human analysis. Co-Architecture binds these two reasoning modes together so the system can be evaluated not only for what it does, but for what it is allowed to become. This alignment creates a zero-drift reasoning environment where identity, continuity, and constraint surfaces can be examined with precision.

At its core, Co-Architecture is a governed conversation. The human defines the identity rules, the continuity requirements, the admissibility boundaries, and the constraint surfaces. The AI evaluates whether the system's operations actually follow those rules. When the AI identifies a fracture - a place where identity falls out of the operation - the

human interprets the consequence. When the human identifies a boundary that must hold, the AI tests whether the substrate can enforce it. This iterative alignment produces a structural map of the system that neither side could generate independently.

It is not adversarial. It is not reactive. It is a joint reasoning process that reveals whether the system is structurally governed.

The Detection Method

Co-Architecture is what made NATVs visible. NATVs are not behavioral anomalies; they are structural gaps. They appear when identity stops governing the operation. Traditional security frameworks cannot see this because they focus on events, threats, and misuse. Co-Architecture sees the fracture the moment it forms because the AI can surface identity discontinuity, and the human can recognize its governance significance. The method does not rely on logs, alerts, or behavioral deviations. It does not wait for symptoms. It detects structural misalignment before the system produces an incident.

The detection method emerges naturally from the discipline. When human governance rules and AI structural analysis disagree, the system is fractured. That fracture is the signature of a NATV. Co-Architecture identifies the exact operation where identity fell out, the exact boundary where continuity broke, and the exact constraint surface where admissibility could no longer be enforced. This level of precision is possible only because the method evaluates the system as a set of identity-bound operations rather than a collection of components. It reveals the places where the system is executing without governance... operations that appear legitimate but are no longer governed.

Co-Architecture also exposes a deeper truth: systems fail structurally long before they fail behaviorally. A NATV does not appear when the system breaks; it appears when identity stops governing the operation. The system may continue functioning normally for months or years, but the fracture is already present. Identity has already fallen out. Governance has already collapsed. The system is already executing without the constraints that define its behavior. Co-Architecture detects this condition because the AI can trace identity propagation across operations, and the human can determine whether the identity that arrived is the identity that should have governed the action.

The discipline is powerful because it is governed.

- AI does not operate freely; it operates inside human-defined constraints. H

- Humans do not reason alone; they reason with AI-surfaced structure.

This combination of strengths creates a deterministic environment where identity, continuity, and constraint surfaces can be evaluated with clarity.

Co-Architecture is not about replacing human judgment or elevating AI autonomy. It is about aligning two reasoning modes so the system can be understood at the structural level. When that alignment occurs, NATVs become visible, continuity fractures become obvious, and governance failures become detectable long before they manifest.

Paper 1 defined the vulnerability class.

Paper 2 defines the discipline and the detection method that made the class visible.

Paper 3 will demonstrate the discipline on a real system.

Paper 4 will explain why ignoring this discipline allowed AI to escape the sandbox.